

HOSSEIN AKHLAGHPOUR

AI Architect | Enterprise AI Platform and Applied GenAI Leader

San Francisco Bay Area, CA | 650-353-3724 | hossein.akhlaghpour@gmail.com | linkedin.com/in/hosseinakhlaghpour

EXECUTIVE SUMMARY

AI architect and machine learning leader with 8+ years of experience delivering production ML systems and 10+ additional years in software architecture, distributed systems, and technical leadership. Defined and deployed enterprise NLP, generative AI, and workflow automation capabilities for legal and CRM products, including natural-language interfaces, information extraction, narrative generation, and secure API orchestration. Earlier work spans large-scale optimization, real-time infrastructure, digital health, genomics, and high-volume data platforms. Combines hands-on ML depth with systems design, executive stakeholder partnership, and cross-functional leadership to deliver secure, scalable enterprise AI platforms. Experienced in setting architecture standards, evaluation and guardrail patterns, and rollout approaches for regulated enterprise workflows.

SELECTED LEADERSHIP HIGHLIGHTS

- Built AI capabilities for enterprise legal and CRM workflows, including a natural-language interface for a CRM product used by 10,000+ customers.
- Defined reusable AI architecture patterns for retrieval, tool invocation, and enterprise workflow orchestration across multiple product surfaces.
- Drove cross-functional architecture decisions that balanced model quality, latency, security, and production reliability for customer-facing AI systems.
- Developed and optimized ML systems operating in real-time environments controlling more than 10,000 batteries.
- Led software and ML initiatives spanning cloud platforms, distributed systems, embedded systems, and high-scale data pipelines.
- Managed engineering teams of up to 10 developers and delivered platforms processing approximately 50 billion records per month.
- Architected customer-facing reporting systems serving more than 6,000 merchants.

CORE STRENGTHS

Enterprise AI Applications | AI System Design | AI/ML Platform Architecture | Reference Architectures | Generative AI & LLM Applications | Model Routing | Evaluation Frameworks | AI Governance & Guardrails | Human-in-the-Loop Workflows | Secure AI Integration | Production Reliability | Workflow Automation | Distributed Systems | Cloud Infrastructure | Technical Leadership

PROFESSIONAL EXPERIENCE

Intapp — Senior Machine Learning Engineer

Palo Alto, CA | Jan 2021 – Present

- Led architecture and delivery of AI-powered NLP and generative AI capabilities for enterprise legal and CRM workflows.
- Built an email signature extraction system using named entity recognition to convert unstructured communications into structured CRM data.
- Developed compliant narrative-generation workflows that used historical notes and user actions to automate legal documentation.
- Created a natural-language interface for a CRM application used by more than 10,000 customers, enabling users to query customized schemas through natural language.
- Engineered secure translation of user questions into API calls, returning both structured UI outputs and natural-language responses.
- Operated in a forward-deployed engineering model with enterprise stakeholders to translate ambiguous legal and CRM workflows into production AI solutions, driving architecture decisions, delivery plans, and adoption.
- Standardized production patterns for model integration, prompt and output controls, and workflow-level validation to improve reliability and governance of enterprise AI features.

AutoGrid — Senior Machine Learning Engineer

Redwood City, CA | Nov 2019 – Jan 2021

- Improved ML systems used to control more than 10,000 batteries in a smart-grid environment.
- Developed scalable ML and optimization pipelines on Kubernetes and Spark to reduce AWS cost while maintaining response-time requirements.
- Enhanced a real-time event dispatcher using greedy heuristic algorithms to improve resolution speed and responsiveness.
- Improved power-outage detection using gradient boosting approaches including XGBoost and CatBoost.
- Developed a vegetation-planning and risk-identification solution using satellite imagery and U-Net segmentation models.

Vitachain — Director of Machine Learning

Belmont, CA | Nov 2017 – Nov 2019

- Led machine-learning development for digital health applications focused on blood-pressure prediction and behavioral recommendations.
- Developed a contextual bandit system using Vowpal Wabbit to recommend optimal user activities based on PPG, Google Fit, and Apple Health data.
- Built a seq2seq LSTM-based model to predict blood pressure from smartwatch PPG signals.

Genapsys — Software Lead

Redwood City, CA | Apr 2013 – Aug 2016

- Developed DNA-sequencing denoising and base-calling algorithms using PCA, Markov models, Naive Bayes, and AdaBoost.
- Led software development across embedded systems, backend services, and cloud-based distributed infrastructure supporting sequencing workflows.
- Built end-to-end systems using Docker, REST services, AWS, Kubernetes, Spark, HBase, Kafka, RabbitMQ, C++, Java, and Spring-based components.

Velti — Engineering Manager

San Francisco, CA | Nov 2010 – Apr 2013

- Led a team of 10 software developers and delivered a large-scale ad analytics and conversion tracking platform.
- Supported high-volume data processing architecture handling approximately 50 billion records per month across HBase, Solr, Amazon RDS, and DynamoDB.

eBay — Senior Software Architect

Brisbane, CA | Jun 2009 – Nov 2010

- Led development of a merchant reporting platform providing daily reports to more than 6,000 merchants.
- Built reporting and scheduling capabilities on top of BIRT and Quartz while maintaining system performance at scale.

Danisco — Senior Software Engineer

Palo Alto, CA | Jul 2008 – Jun 2009

- Led integration with Google Enterprise Search Appliance through Lucene-based interfaces.
- Supported migration of more than 50,000 documents from IBM content management systems to Alfresco CMS.

TECHNICAL SKILLS

Programming / Platforms: Python, FastAPI, gRPC, REST APIs, Docker, Kubernetes, AWS, Spark, Kafka, Airflow, PostgreSQL, Redis, Vector Databases (Pinecone/FAISS), CI/CD (GitHub Actions/Azure DevOps), MLflow, Kubeflow
Machine Learning / AI: PyTorch, Transformers, LLM Applications, Retrieval-Augmented Generation (RAG), Fine-tuning (LoRA/QLoRA), Diffusion Models, Agentic Workflows (tool use/function calling), Prompt Engineering, Evaluation & Guardrails, Embeddings and Reranking, NLP/NER, XGBoost/CatBoost, U-Net, LSTM, Contextual Bandits

EDUCATION

University of Cincinnati — Master of Science in Physics